

Aram Petrosyan

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EDUCATION

University of California, Los Angeles (UCLA)

Los Angeles, CA

Master of Science in Data Science Engineering – GPA: 4.00

June 2028

- **Relevant Coursework:** Machine Learning, Neural Networks, Automated Reasoning, Database Systems

University of California, Los Angeles (UCLA)

Los Angeles, CA

Bachelor of Science in Computer Science – GPA: 3.86

June 2026

- **Relevant Coursework:** Data Structures, Computer Architecture, Operating Systems, Software Construction
- **Honors:** Dean's Honors List

EXPERIENCE

Software Engineering Intern

June 2026 – Present

IP Law Firm

Los Angeles, CA

- Co-developed a Python/FastAPI platform using OpenAI, cutting hours of Office Action analysis and strategy drafting.
- Automated patent parsing, OCR, and Outlook drafting, saving up to 1 hour of manual review and correspondence per case.
- Built secure attorney-in-the-loop workflows with input validation, protected credentials, and ephemeral case storage.
- Streamlined deployment and testing to support reliable use across the firm's Citrix environment.

Software Engineering Intern

Jan 2025 – April 2025

ITJ Group

Los Angeles, CA

- Owned SEO strategy, implementing schema markup, redirect rules, and request optimizations increasing organic traffic 45%.
- Diagnosed site performance bottlenecks by analyzing internal API calls, network requests, and component structure.
- Improved page load times by 30% through targeted frontend, API request, and component-level optimizations.

Freelance Web Developer

Feb 2024 – Present

Self-Employed

Los Angeles, CA

- Designed production WordPress websites for small businesses, handling development, hosting, SEO, and maintenance.
- Built a legal-services website generating an average of 3 inbound leads per day through SEO and conversion-focused design.
- Developed a custom jewelry business website that contributed to acquiring 10+ long-term clients.

PROJECTS

XR Binary Analysis

- Built a Python static-analysis pipeline to identify security vulnerabilities in compiled Unity IL2CPP applications.
- Automated detection of plaintext logging, insecure HTTP usage, and sensitive camera/microphone access patterns.
- Integrated Il2CppDumper output with custom string and function analysis to systematically inspect compiled XR binaries.
- Reduced manual security-review time by 80% through automated vulnerability detection and analysis.

FairShare App

- Developed expense-entry, real-time balance-tracking, and settlement interfaces for a full-stack group expense application.
- Built frontend workflows for AI-powered receipt scanning, enabling users to submit expenses from receipt photos.
- Integrated calculated expense splits with team-built OpenAI receipt extraction and Twilio SMS functionality.
- Collaborated on a 6-person Agile team using shared APIs and modular components to deliver the application.

FPGA Craps Game | Verilog, Vivado, Basys3 Board

- Built a fully hardware-driven craps game on a Basys3 FPGA using Verilog and Vivado.
- Designed the finite-state machine and datapath controlling game states, dice rolls, scoring, and win/loss conditions.
- Implemented a pseudo-random number generator and seven-segment display controller for real-time gameplay.
- Debugged timing constraints and propagation delays across modules to improve signal reliability and system behavior.

TCP Handshake

- Implemented the TCP three-way handshake from scratch using C socket programming on Linux.
- Constructed and parsed SYN, SYN-ACK, and ACK segments to simulate client-server connection establishment.
- Analyzed sequence numbers, acknowledgments, retransmission behavior, and packet exchanges at the transport layer.
- Validated reliable connection establishment by inspecting and debugging TCP segment behavior.

Nova Search Engine

- Engineered a full-stack search engine with an asynchronous Python crawler, FastAPI backend, and Next.js frontend.
- Crawled 10,000+ sites in under 90 seconds using aiohttp and BeautifulSoup with TF-IDF indexing and tokenization.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Typescript/JavaScript, HTML/CSS, R

Cloud Infrastructure: AWS (Lambda, API Gateway), Docker, Linux, Git

Frameworks & Data: FastAPI, Node.js, React, Next.js, MongoDB, Firebase, PyTorch

Networking & Security: TCP/IP, Socket Programming, REST APIs, Ghidra

Certifications: AWS Cloud Practitioner